Institut Universitaire des sciences

FST

Rapport du travail de Laboratoire #3

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Date : 3 Novembre 2026

L'objectif de ce TD est de :

1. Savoir attribuer des adresses IP valides aux machines.
2. Comprendre l'adressage IPv4 et IPv6.

3. Configurer des adresses IP sur des hôtes et routeurs dans Cisco Packet Tracer.
4. Vérifier la connectivité avec les commandes ping et ping

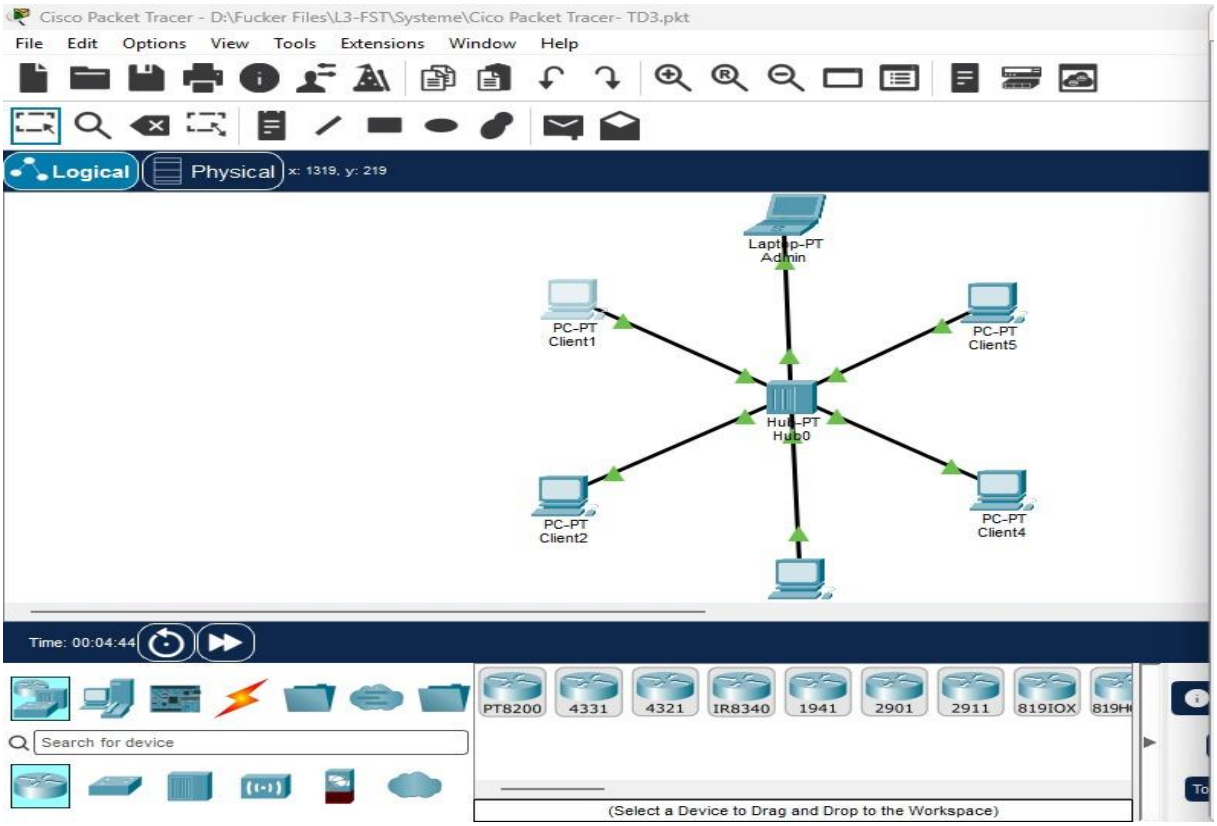
Les démarches suivies.

1. Construction du topologie de sisco parket Tracer.
2. Configuration IPv4 et Configuration.
3. Configuration IPv6 et Configuration.

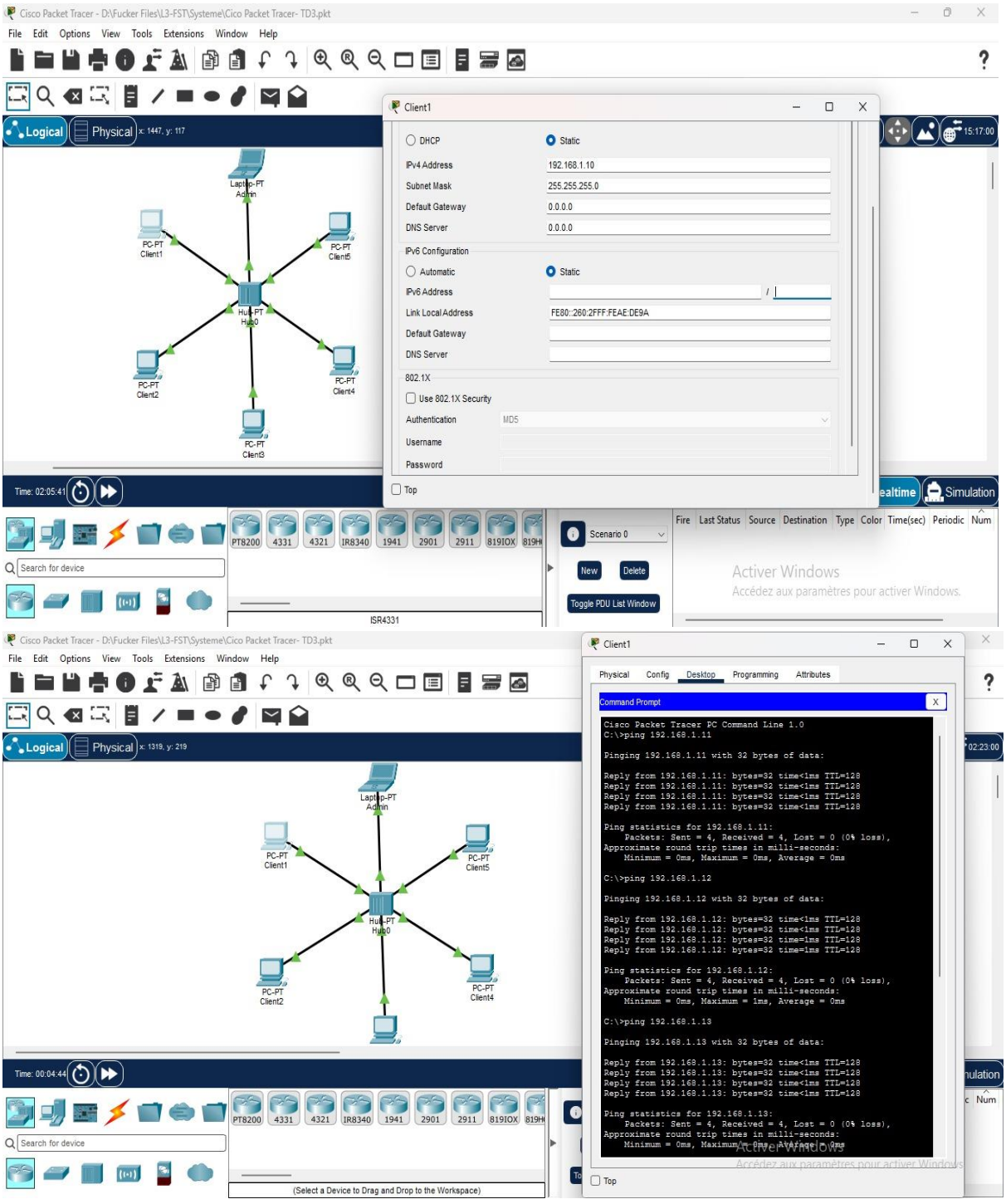
Étape 1.

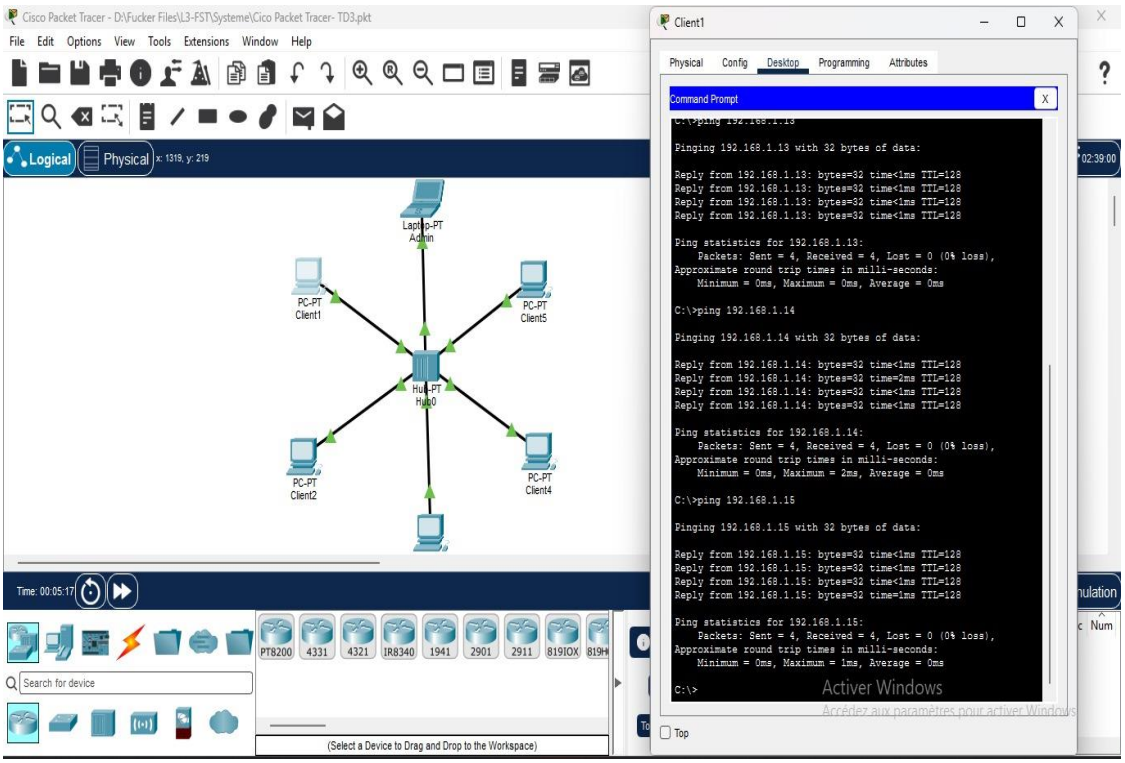
- Topologie en étoile

Nou allons configurer cette topologie en IPv4 et vérifier les connectivités.

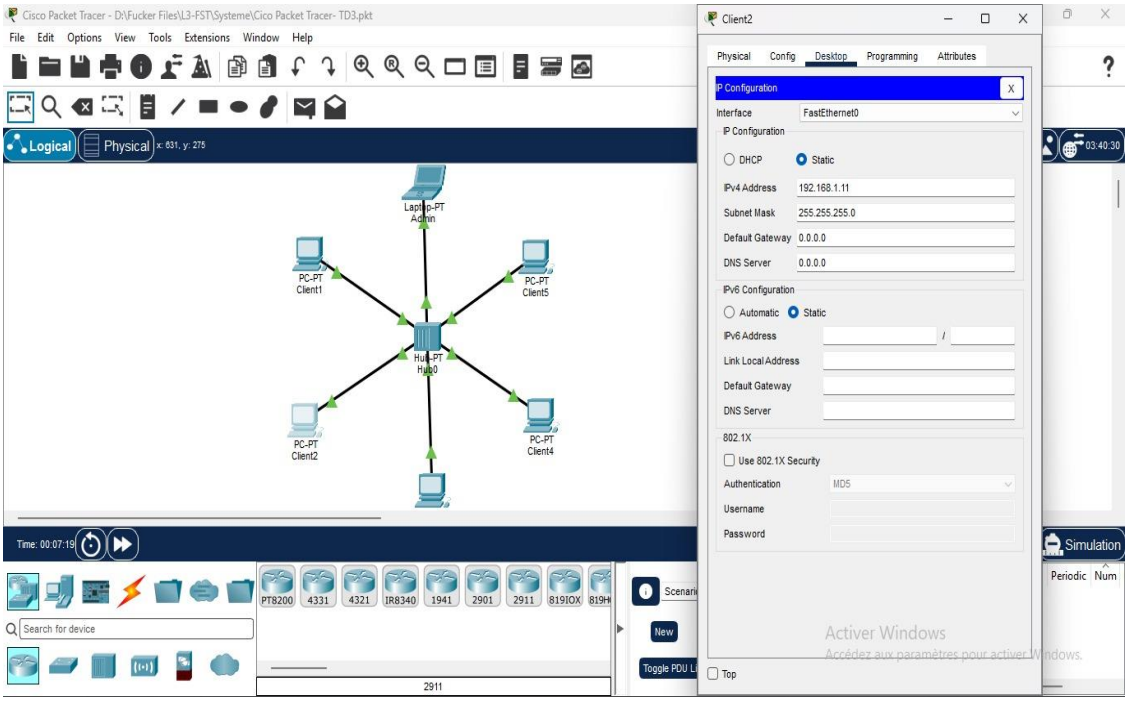


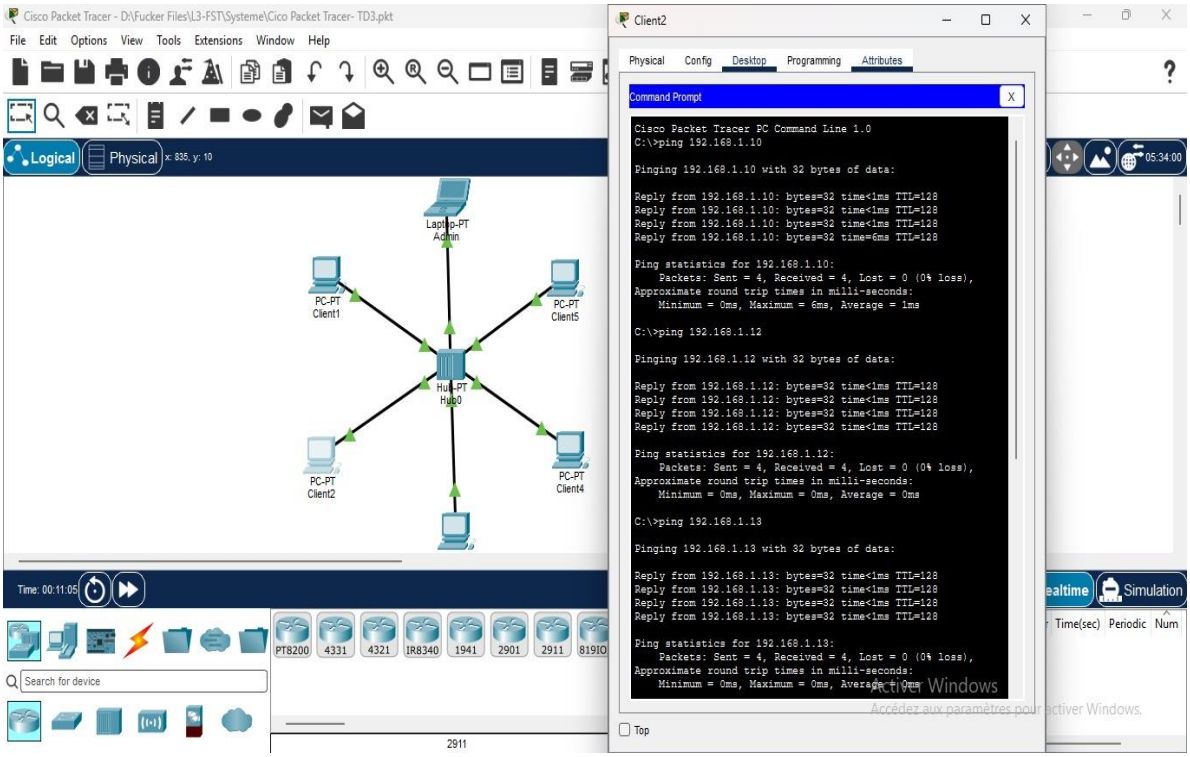
1. Configuration et vérification du client 1 en IPv4.


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1447, y: 117
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 02:05:41
Client1
DHCP
Static
IPv4 Address 192.168.1.10
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::260:2FFF:FEAE:DE9A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Simulation
PT8200 4331 4321 IR8340 1941 2901 2911 819IOX
Search for device
ISR4331
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
Activer Windows
Accédez aux paramètres pour activer Windows.
Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1319, y: 219
Time: 00:04:44
Client1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top
(Select a Device to Drag and Drop to the Workspace)

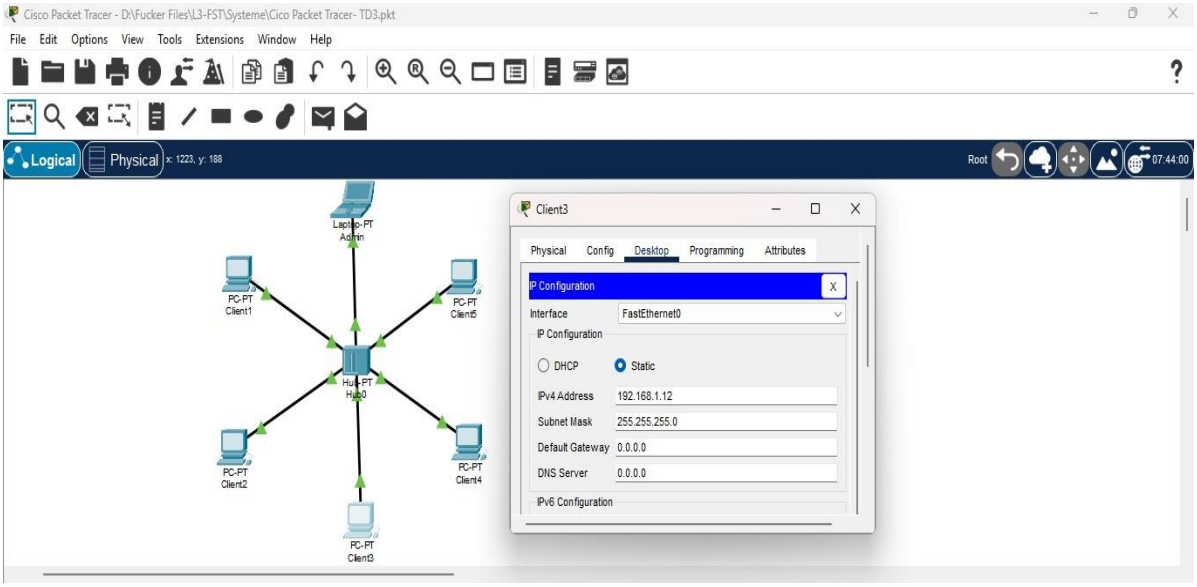


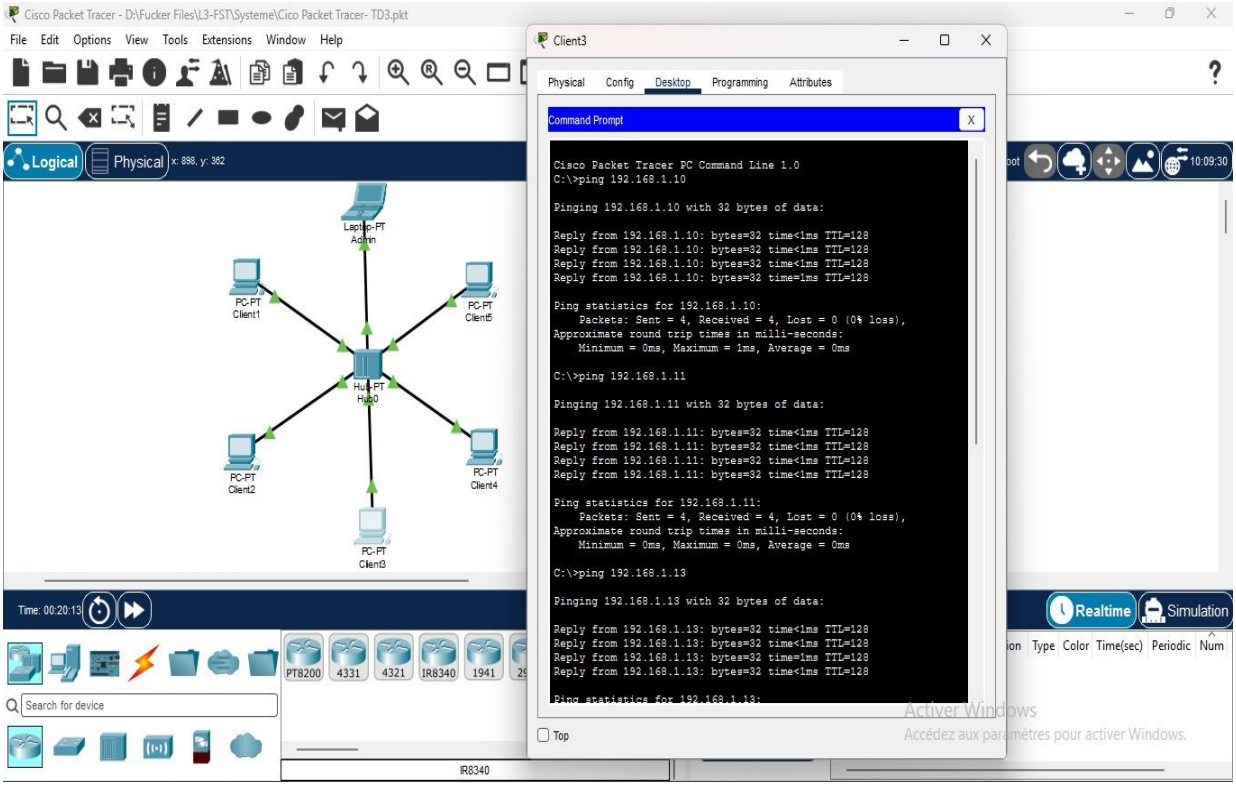
2- Configuration et vérification client 2 IPv4.

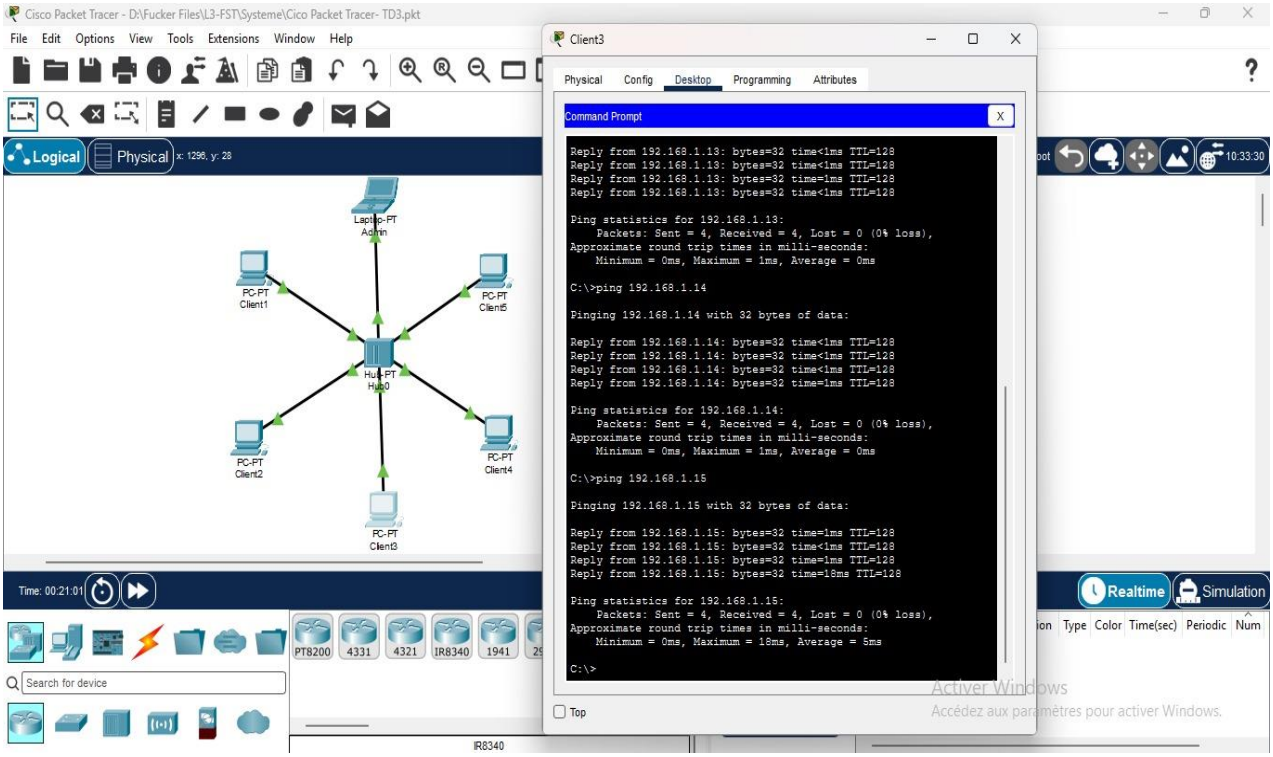




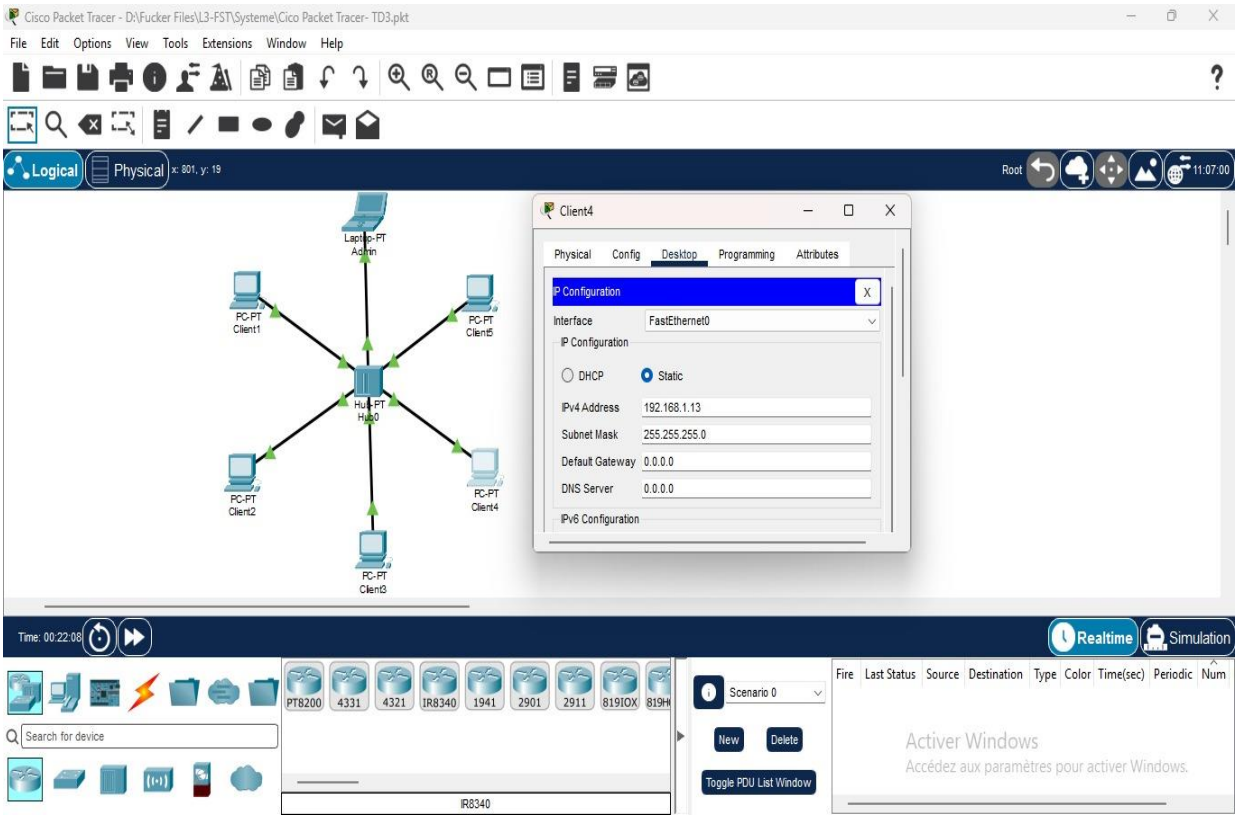
3- Configuration et verification du Client3 en IPv4 :

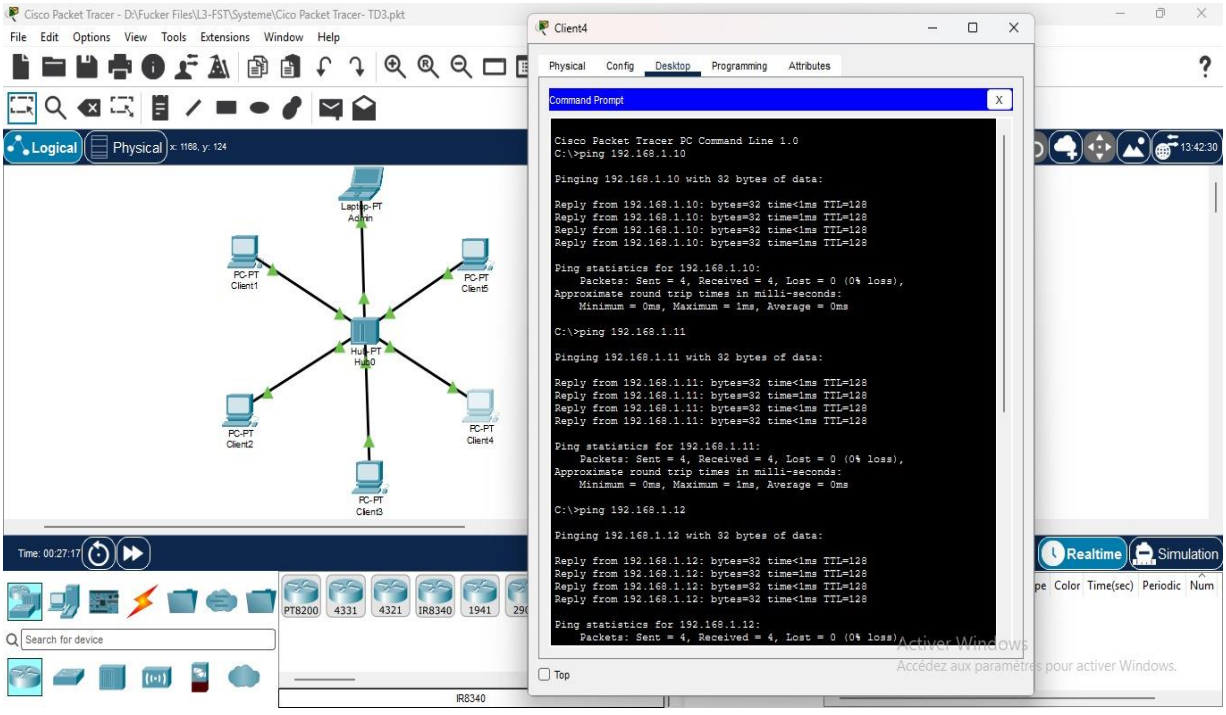


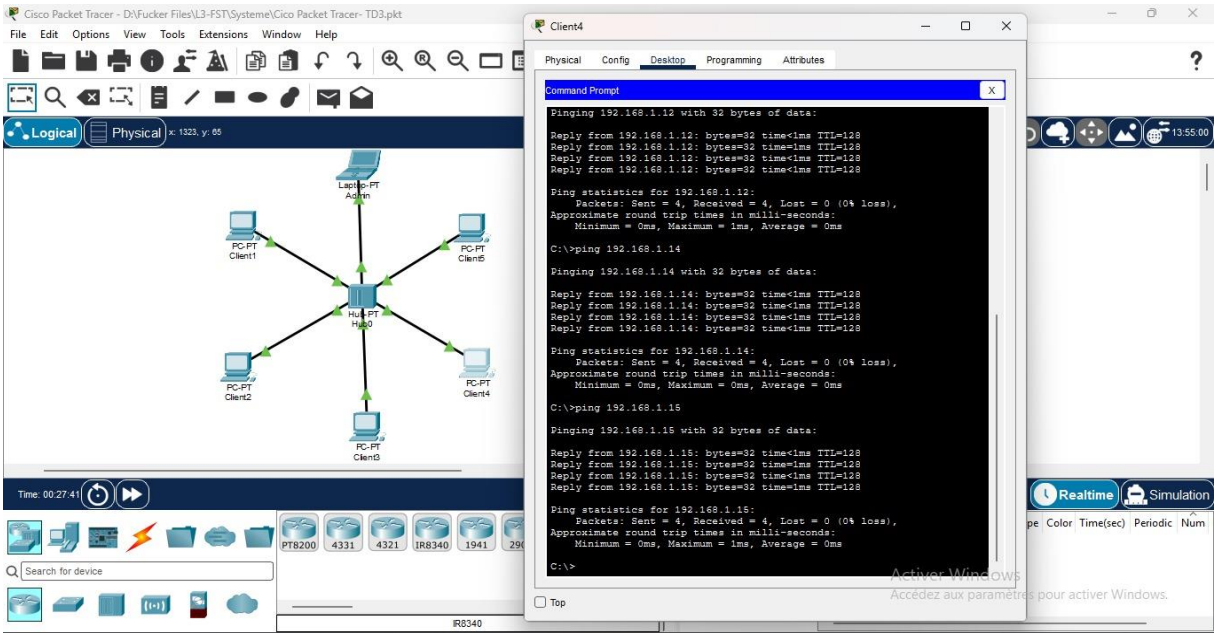




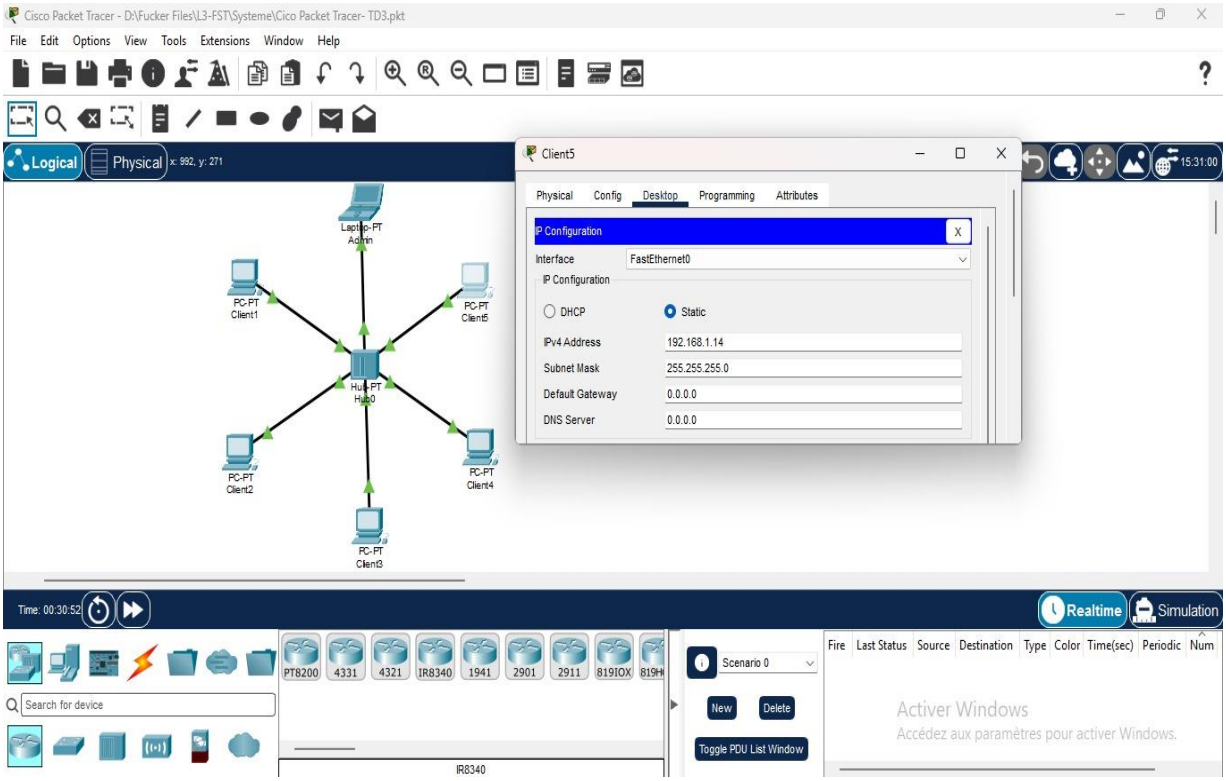
4- Configuration et verification du Client4 en IPv4 :

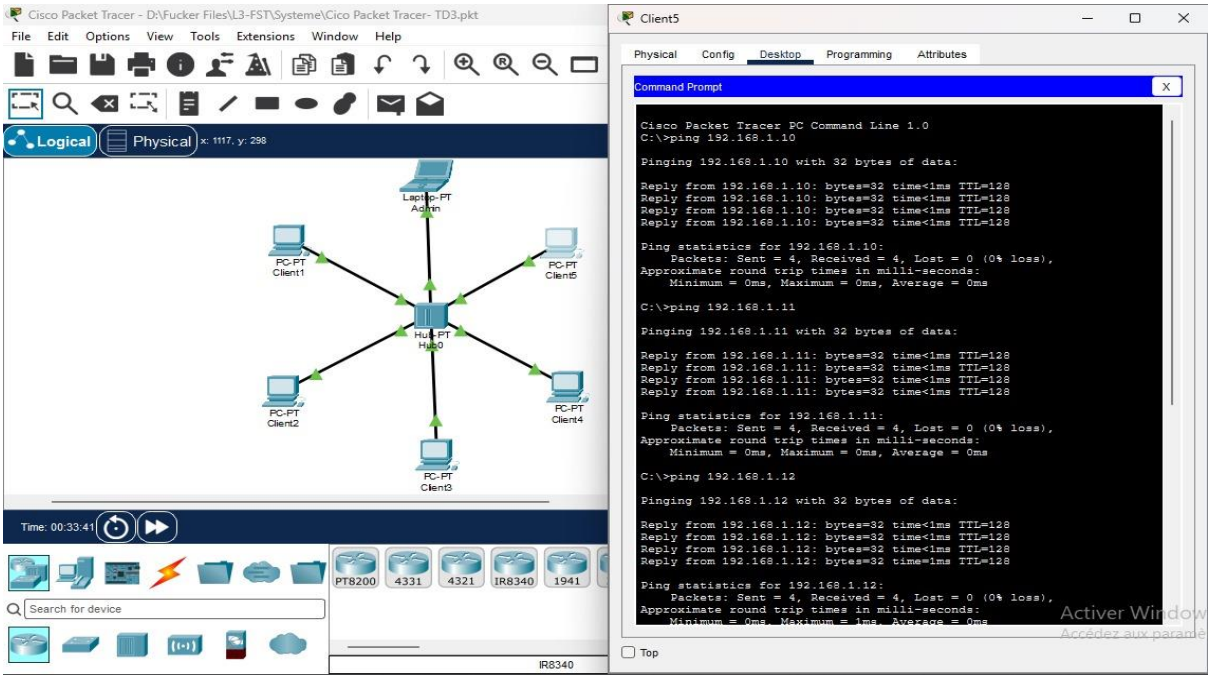

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 801, y: 19
Root 11:07:00
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client4
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
IP Configuration
DHCP Static
IPv4 Address 192.168.1.13
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Time: 00:22:08
Realtime Simulation
PT8200 4331 4321 IR8340 1941 2901 2911 819IOX
Search for device
IR8340
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
Activer Windows
Accédez aux paramètres pour activer Windows.

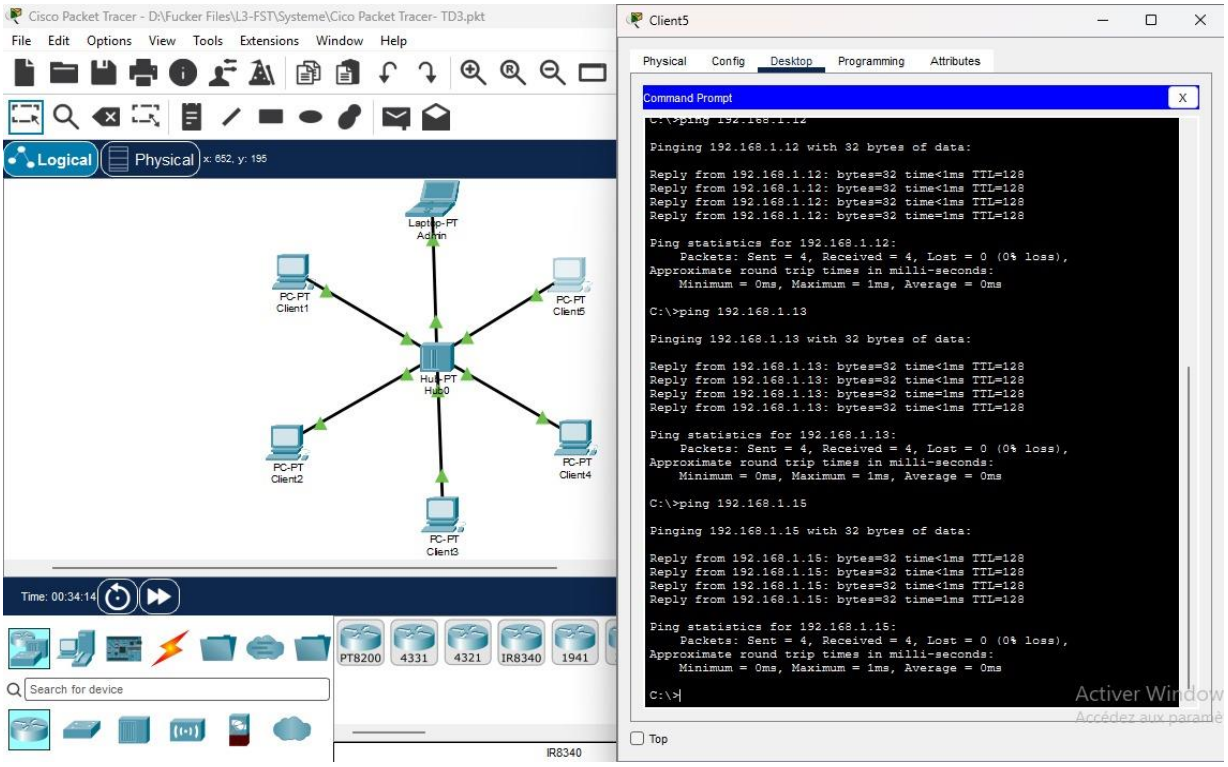

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1168, y: 124
13:42:30
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client4
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Top
Time: 00:27:17
Realtime Simulation
PT8200 4331 4321 IR8340 1941
Search for device
IR8340
Activer Windows
Accédez aux paramètres pour activer Windows.



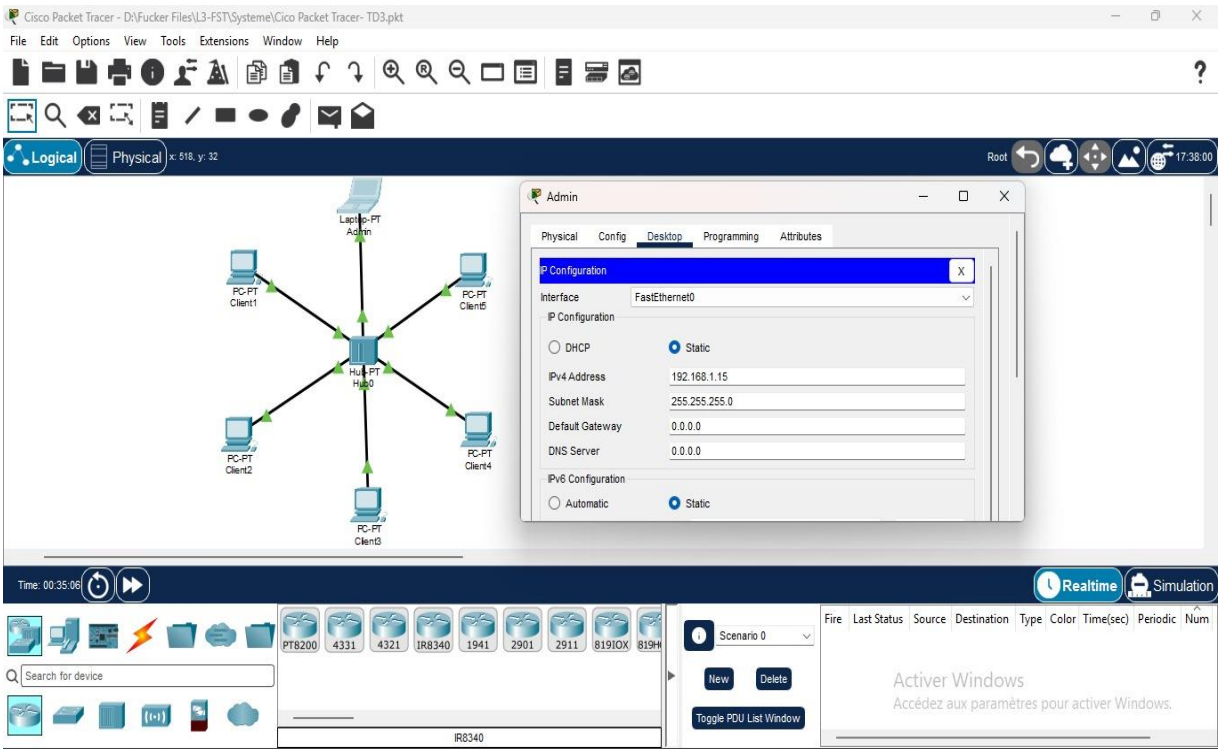
5- Configuration et verification du Client5 en IPv4 :

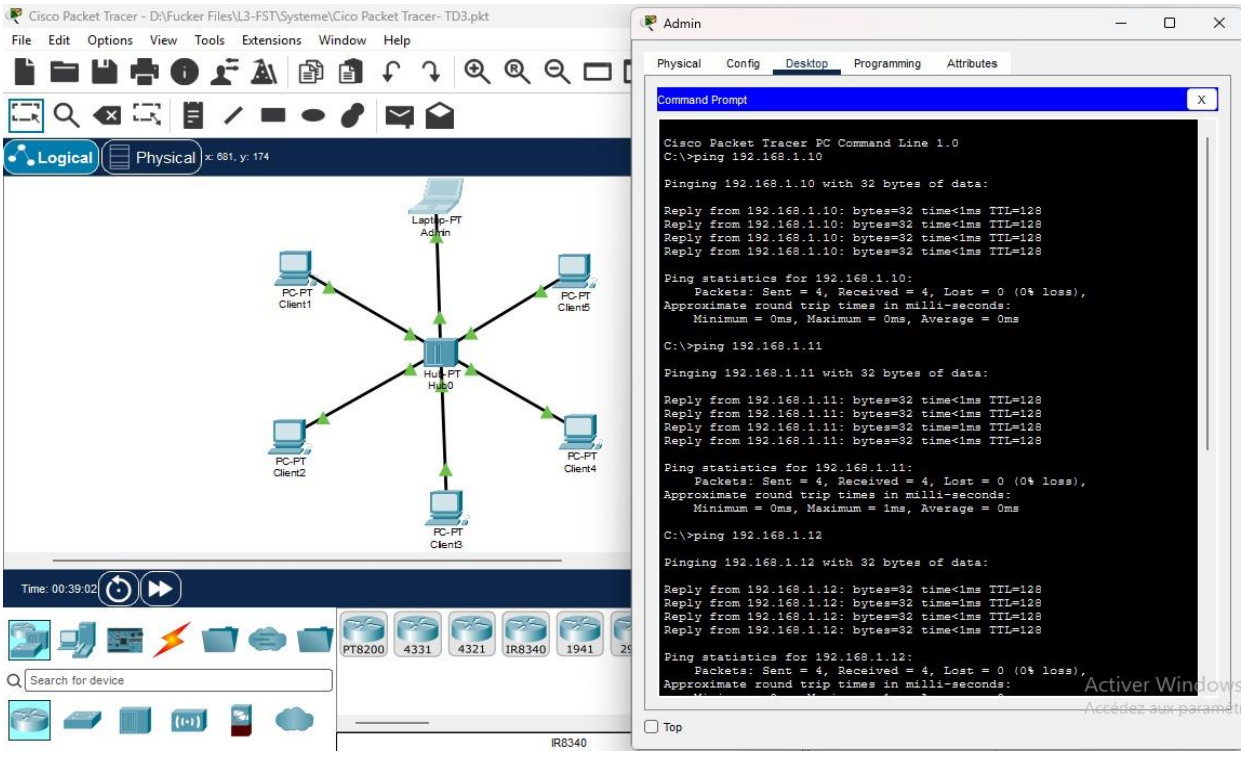


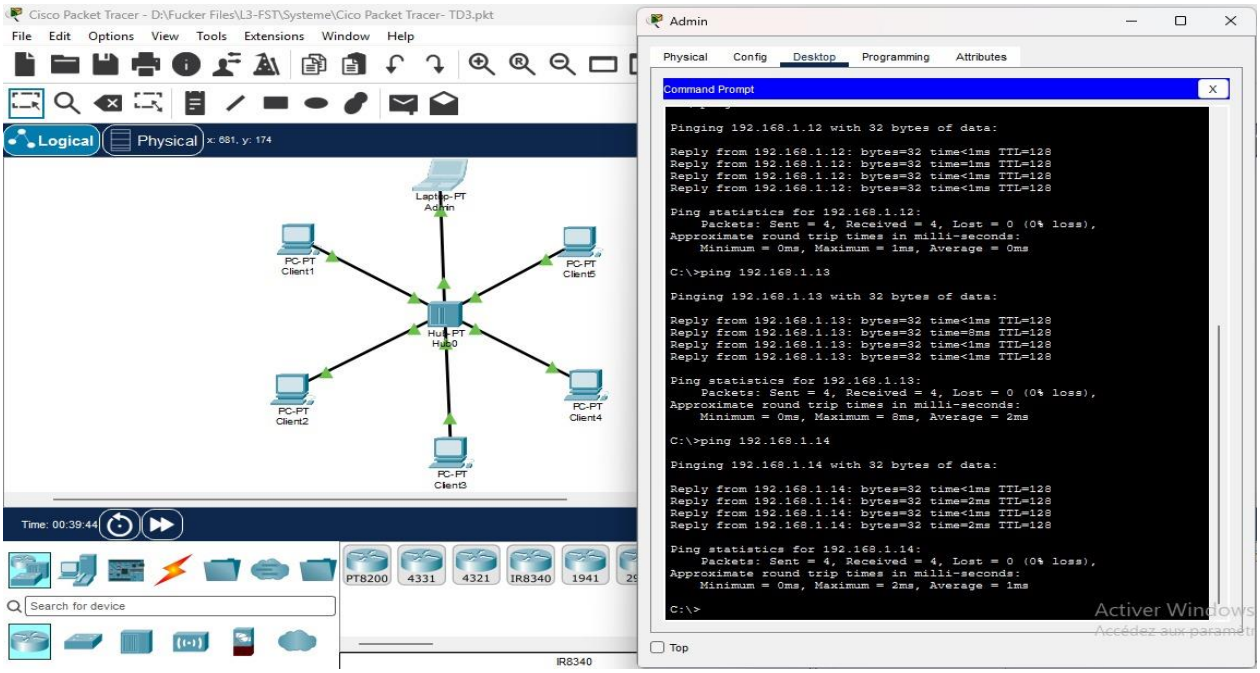




6- Configuration et verification du Laptop Admin en IPv4 :


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Admin
Physical Config Desktop Programming Attributes
IP Configuration
Interface FastEthernet0
DHCP
Static
IPv4 Address 192.168.1.15
Subnet Mask 255.255.255.0
Default Gateway 0.0.0.0
DNS Server 0.0.0.0
IPv6 Configuration
Automatic
Static
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:35:06
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num
Activer Windows
Accédez aux paramètres pour activer Windows.
Search for device
IR8340

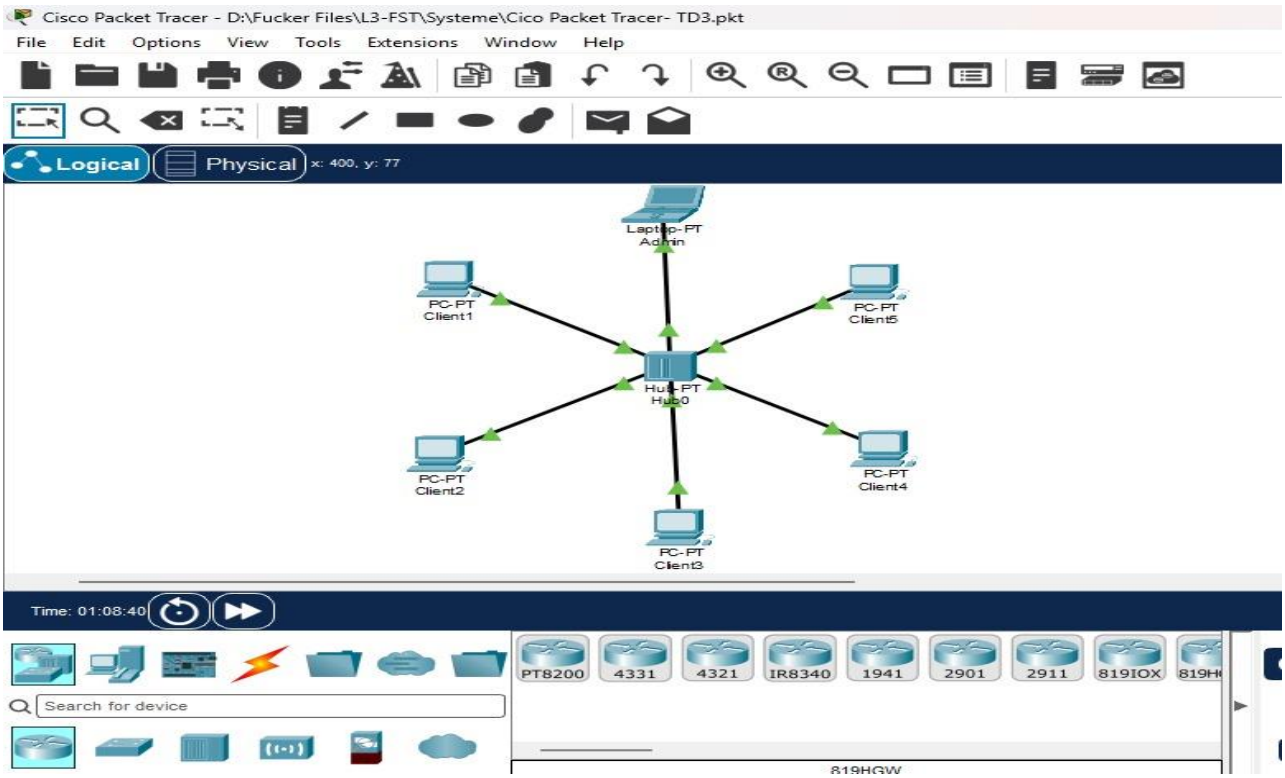

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Admin
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10
Pinging 192.168.1.10 with 32 bytes of data:
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.11
Pinging 192.168.1.11 with 32 bytes of data:
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.11:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.1.12
Pinging 192.168.1.12 with 32 bytes of data:
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Reply from 192.168.1.12: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.12:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Top
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:39:02
Search for device
IR8340
Activer Windows



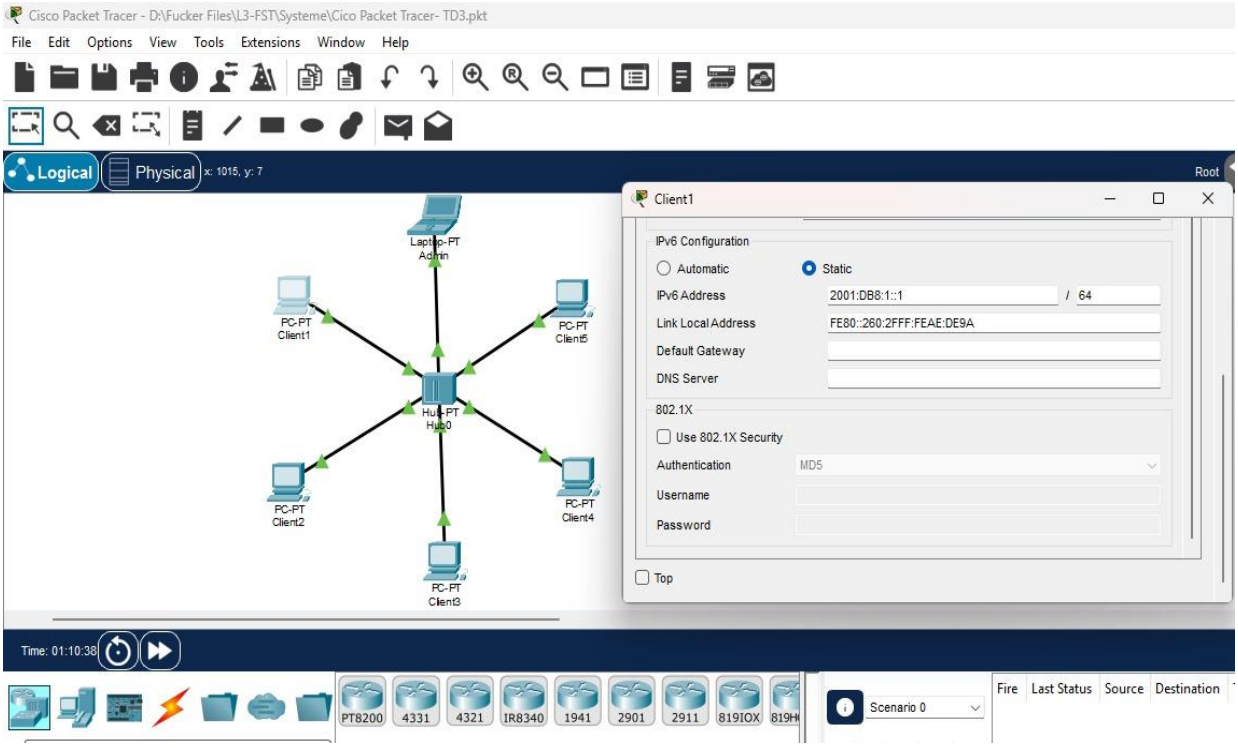
Étape 2

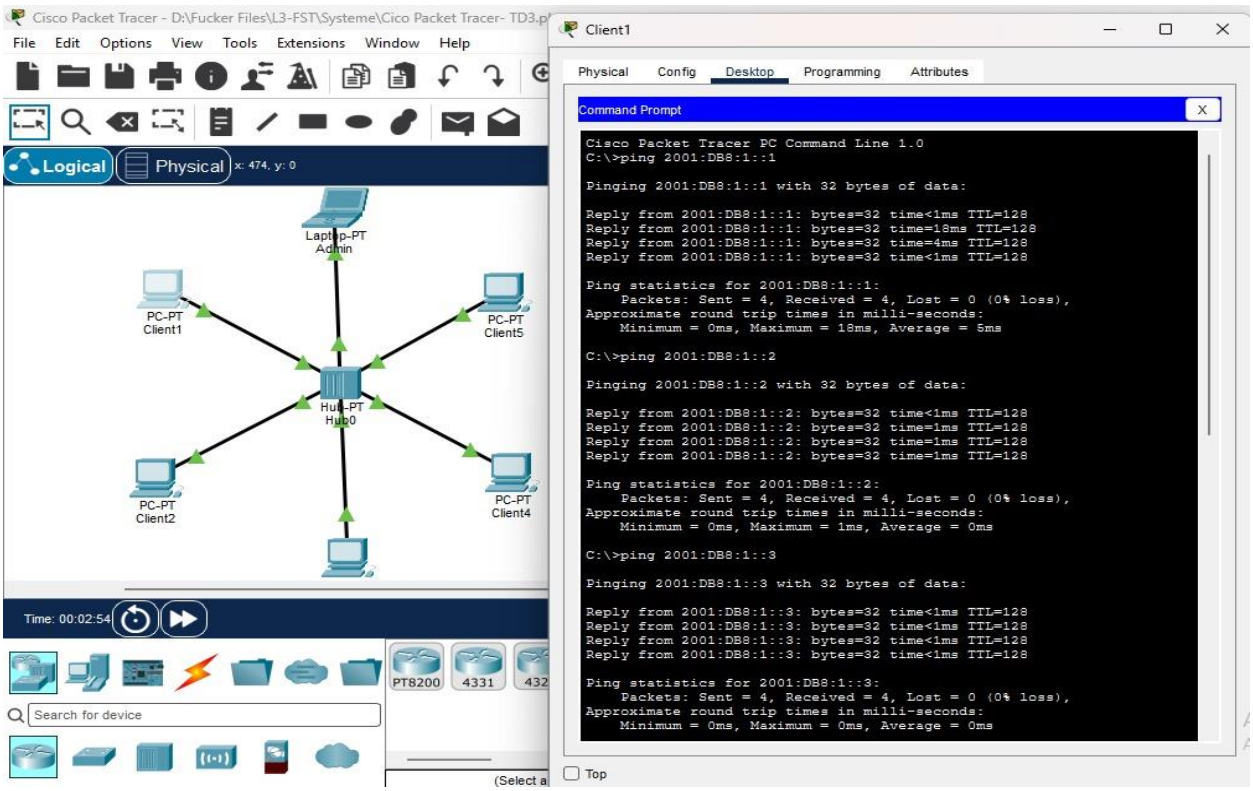
On a fait le même choix du Topologie en Étoile de : 5 pc, 1 laptop et un

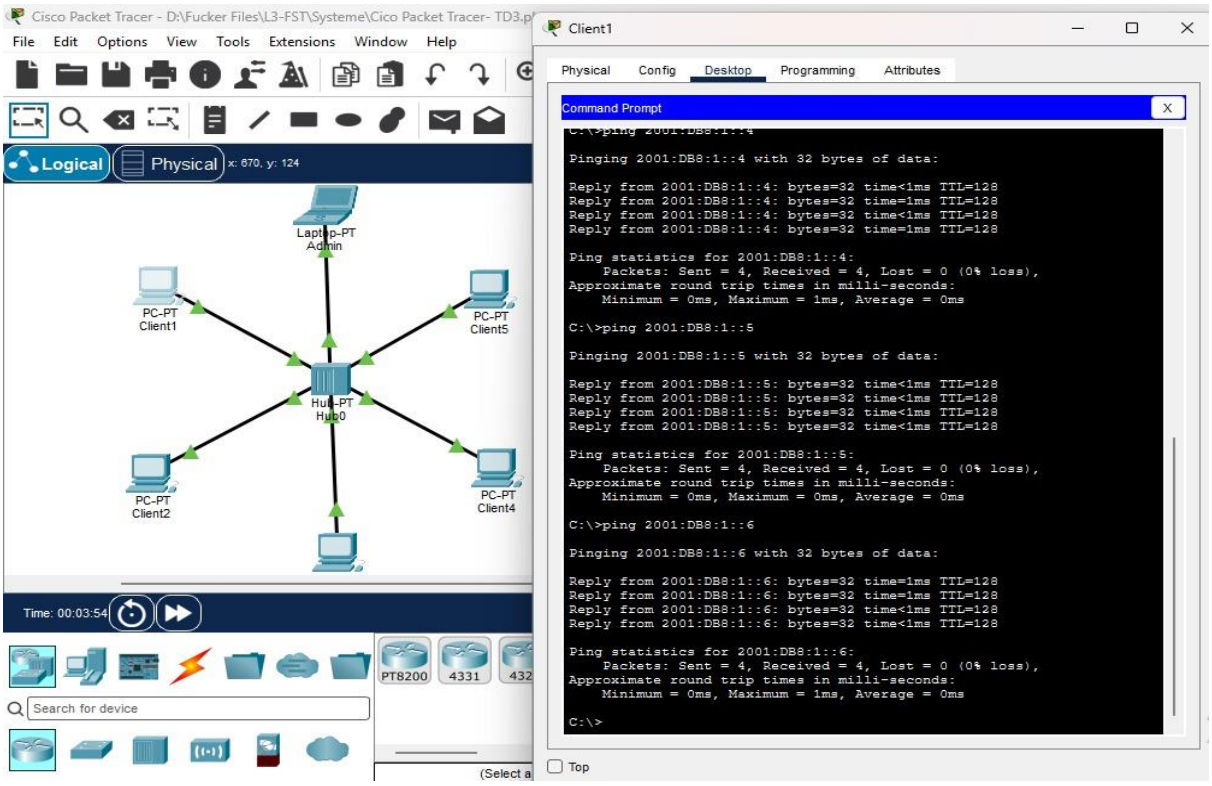
Hub. Pour la configuration et la verification des connectivité en IPv6.


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 400, y: 77
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 01:08:40
Search for device
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
819HGW

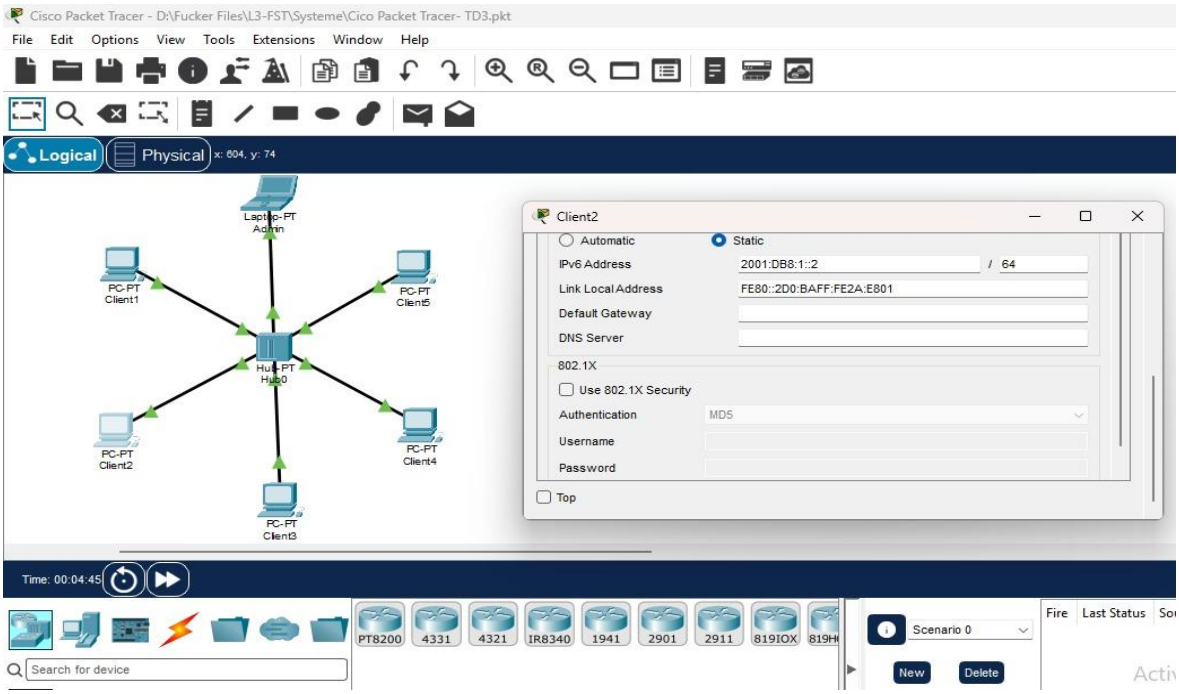
Client 1.

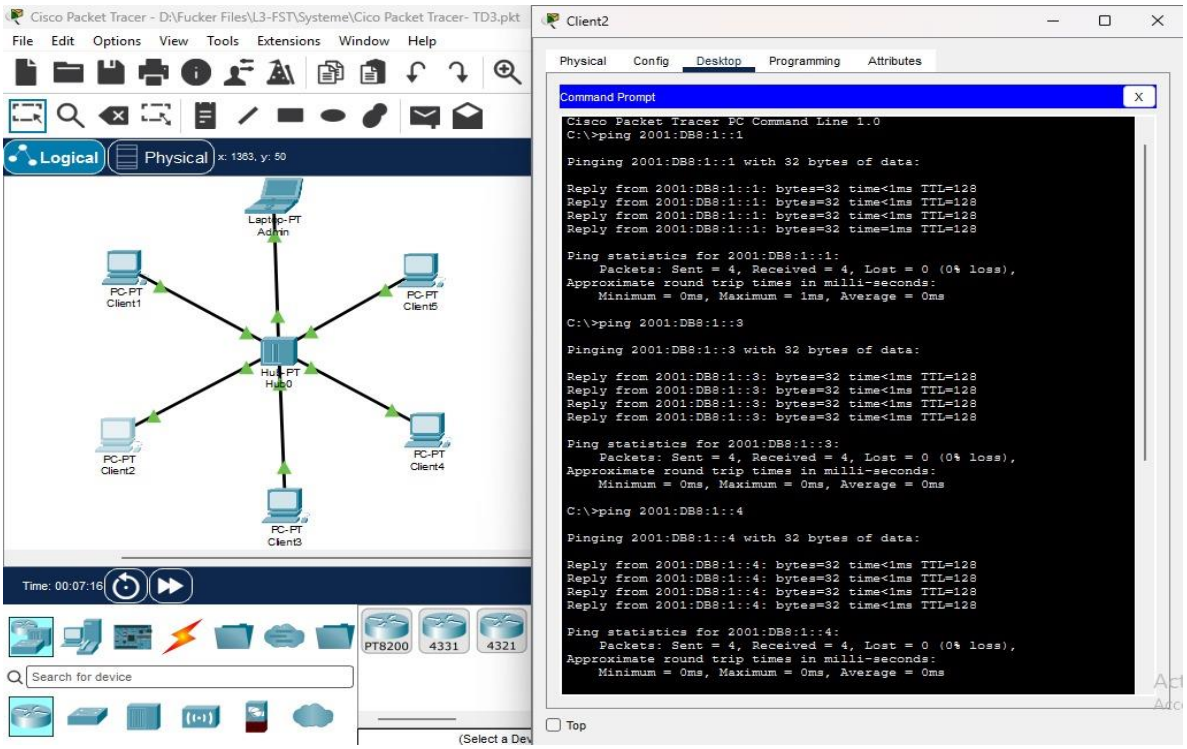

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1015, y: 7
Root
Client1
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::1
/ 64
Link Local Address
FE80::260:2FFF:FEAE:DE9A
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 01:10:38
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
Scenario 0
Fire
Last Status
Source
Destination

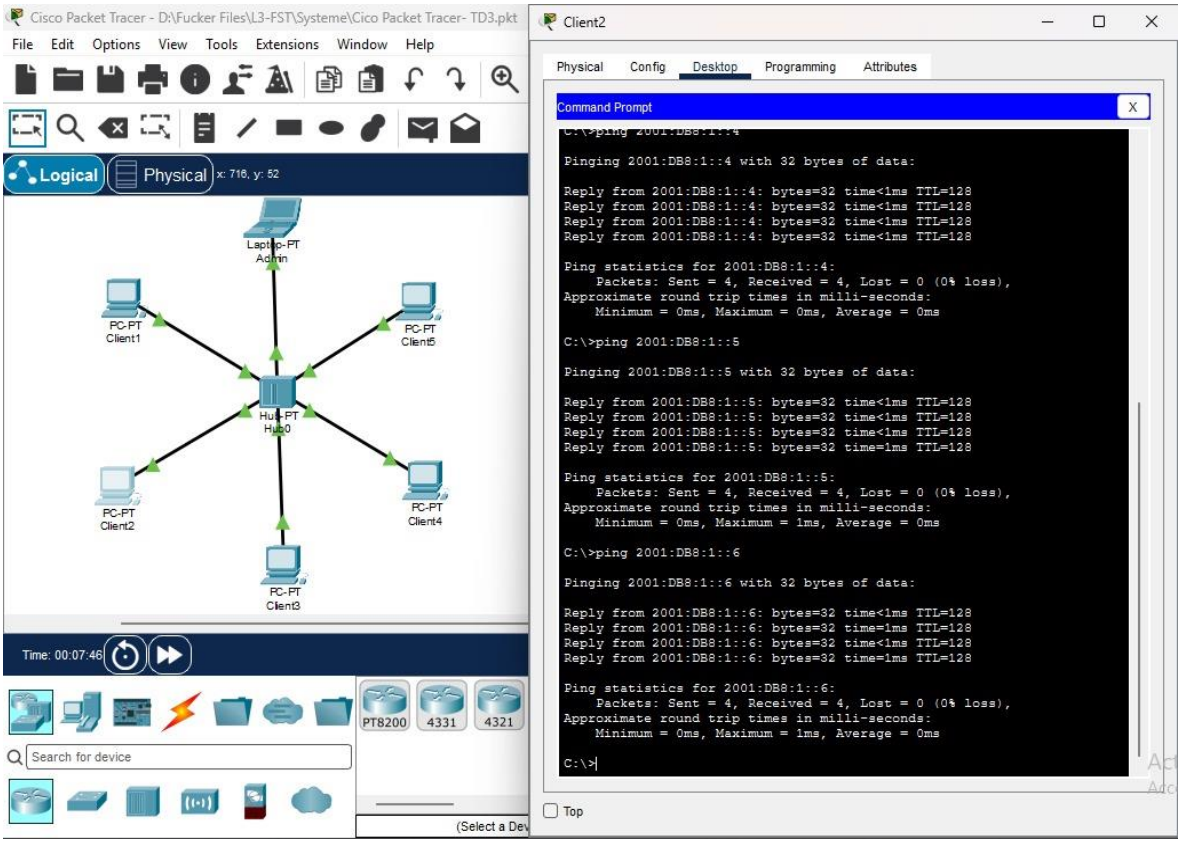

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.p
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 474, y: 0
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
Time: 00:02:54
Search for device
PT8200
4331
Client1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=18ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 18ms, Average = 5ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.p
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 870, y: 124
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
Time: 00:03:54
Search for device
PT8200
4331
Client1
Physical Config Desktop Programming Attributes
Command Prompt
Pinging 2001:DB8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::5
Pinging 2001:DB8:1::5 with 32 bytes of data:
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::6
Pinging 2001:DB8:1::6 with 32 bytes of data:
Reply from 2001:DB8:1::6: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

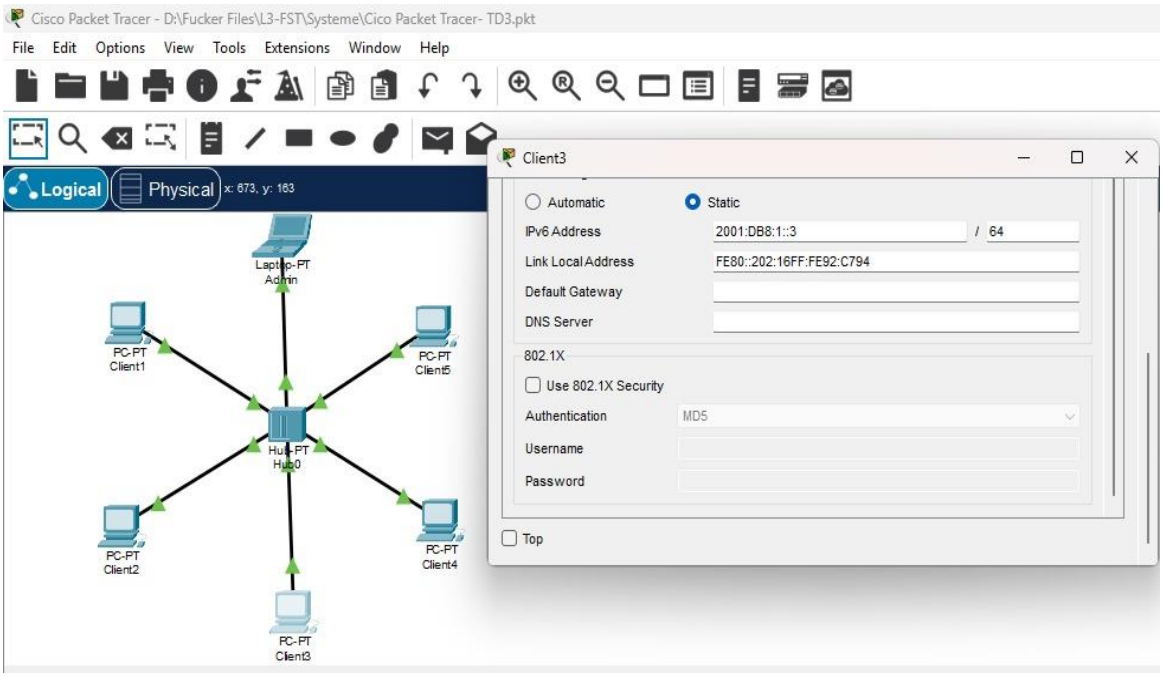
Client 2.

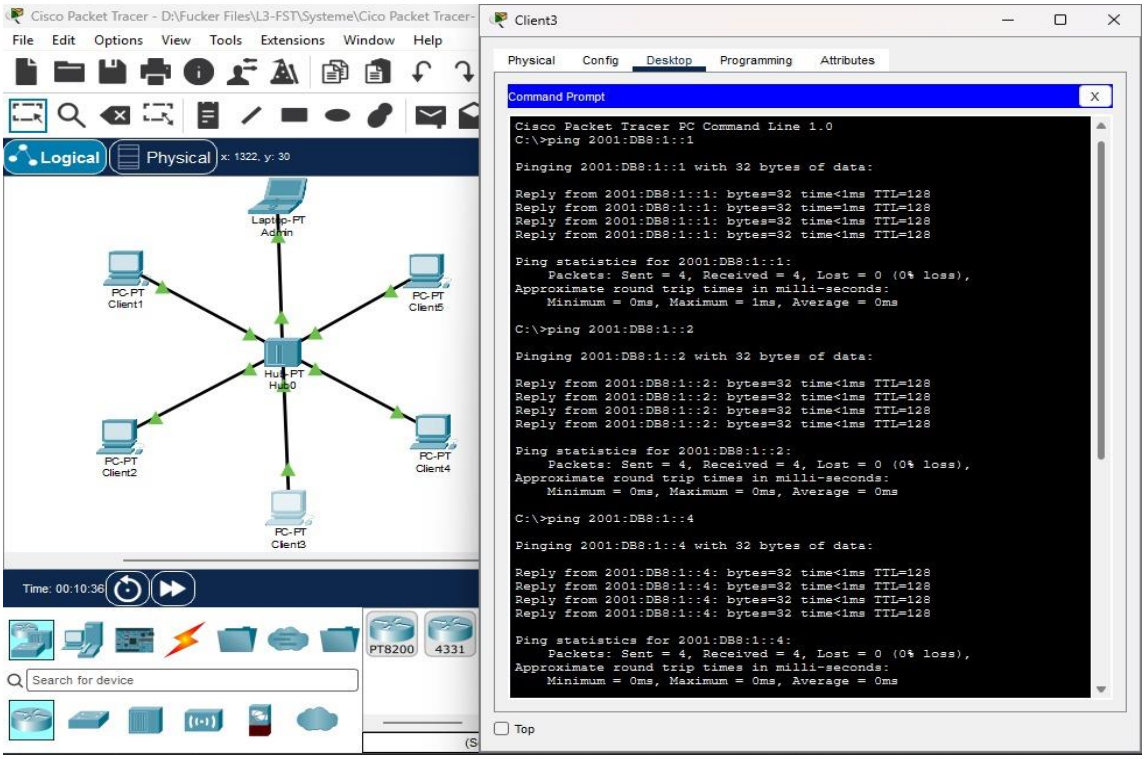

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 604, y: 74
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client2
Automatic
Static
IPv6 Address
2001:DB8:1::2
/ 64
Link Local Address
FE80::2D0:BAFF:FE2A:E801
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Time: 00:04:45
PT8200
4331
4321
IR8340
1941
2901
2911
819IOX
Search for device
Scenario 0
New
Delete
Fire
Last Status

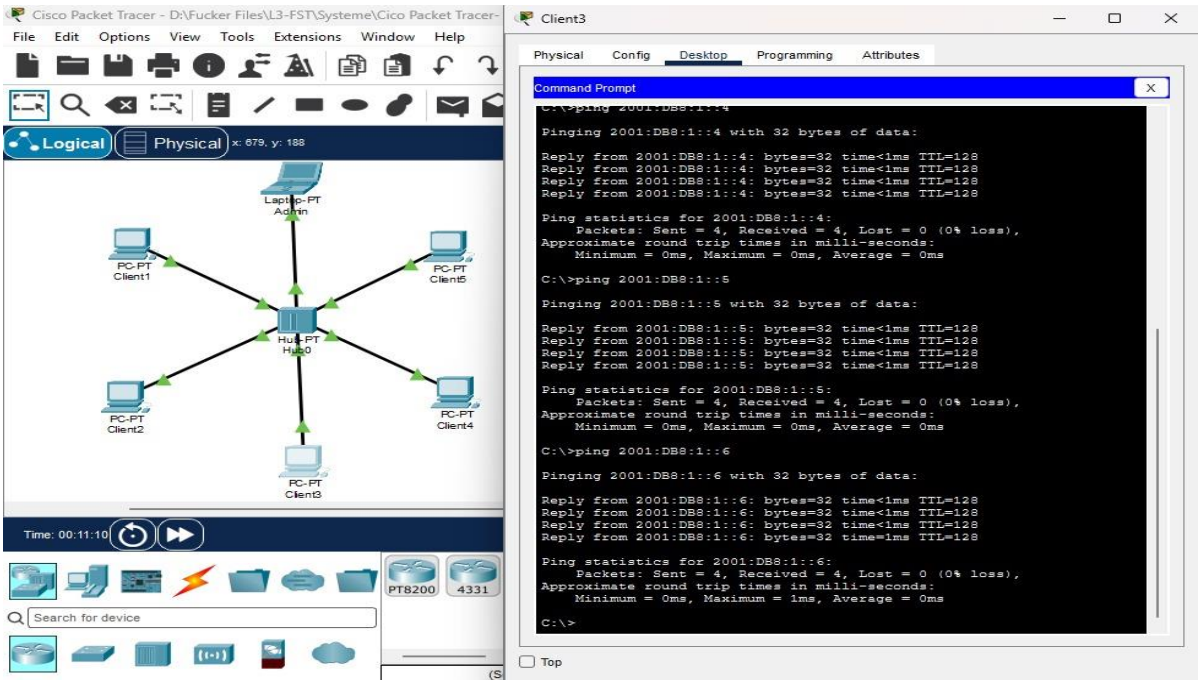

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1383, y: 50
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:07:16
PT8200
4331
4321
Search for device
(Select a Dev
Client2
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::4
Pinging 2001:DB8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 716, y: 52
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:07:46
PT8200 4331 4321
Search for device
Client2
Physical Config Desktop Programming Attributes
Command Prompt
Pinging 2001:DB8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::5
Pinging 2001:DB8:1::5 with 32 bytes of data:
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::6
Pinging 2001:DB8:1::6 with 32 bytes of data:
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

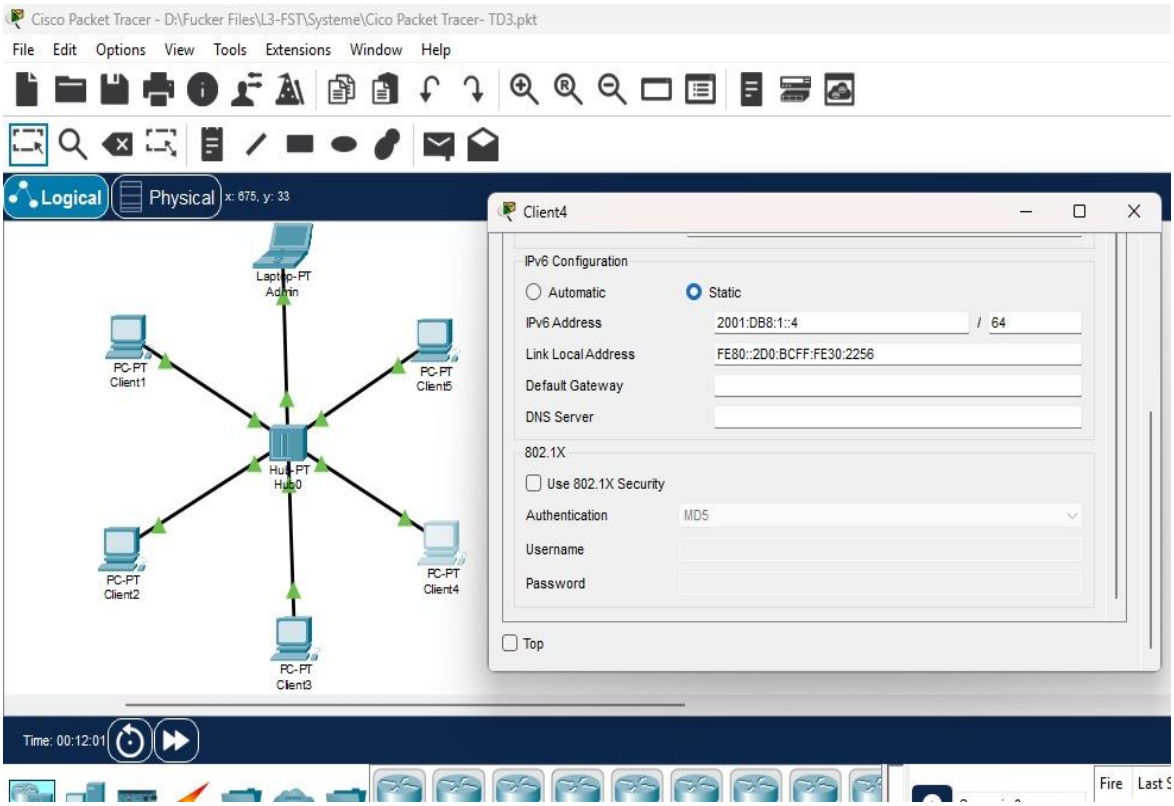
Client 3.

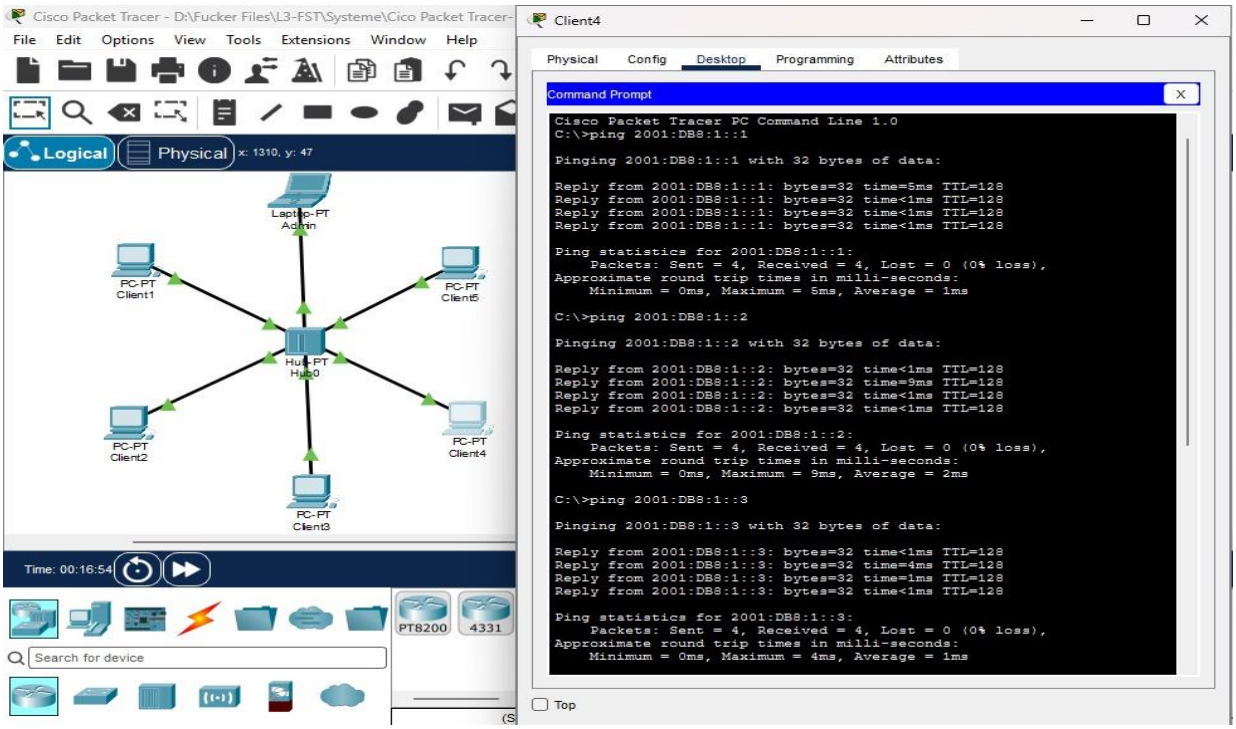

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 673, y: 183
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client3
Automatic
Static
IPv6 Address 2001:DB8:1::3 / 64
Link Local Address FE80::202:16FF:FE92:C794
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top

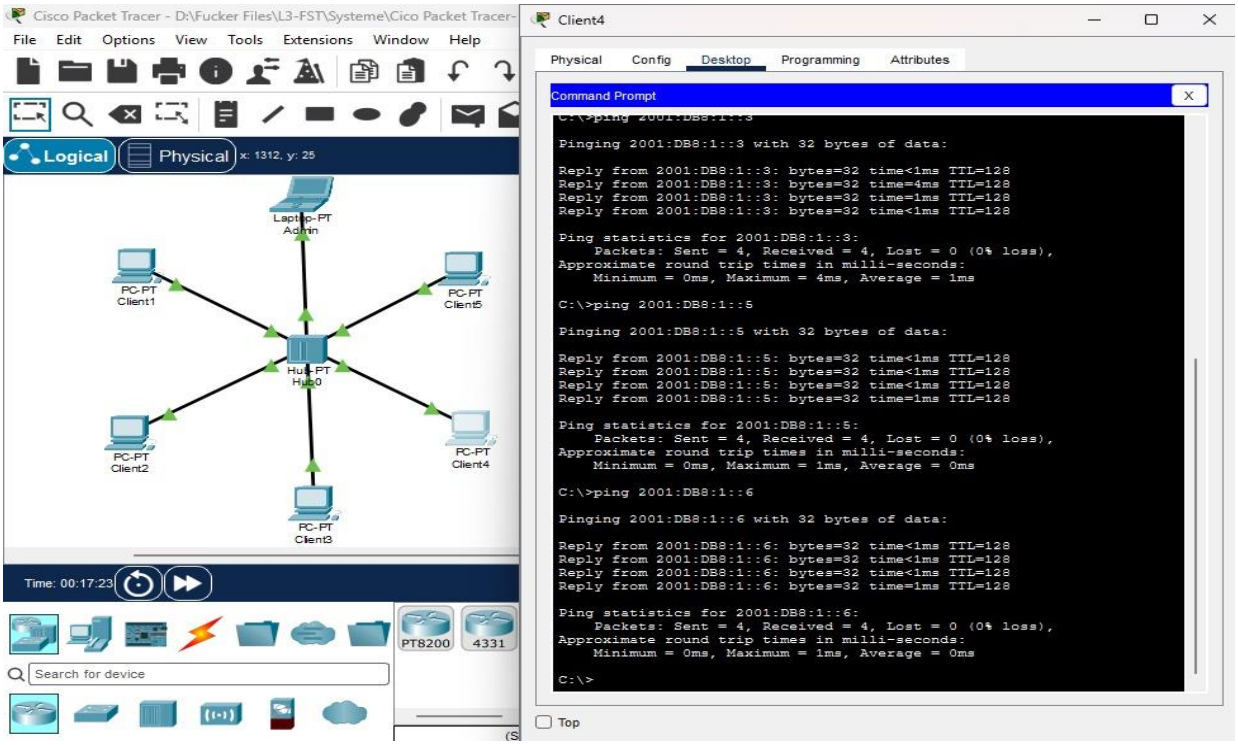

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer-
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1322, y: 30
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:10:36
PT8200
4331
Search for device
Client3
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::4
Pinging 2001:DB8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer-
File Edit Options View Tools Extensions Window Help
Logical Physical x: 679, y: 188
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:11:10
PT8200
4331
Search for device
Client3
Physical Config Desktop Programming Attributes
Command Prompt
Pinging 2001:DB8:1::4 with 32 bytes of data:
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::4: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::5
Pinging 2001:DB8:1::5 with 32 bytes of data:
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 2001:DB8:1::6
Pinging 2001:DB8:1::6 with 32 bytes of data:
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

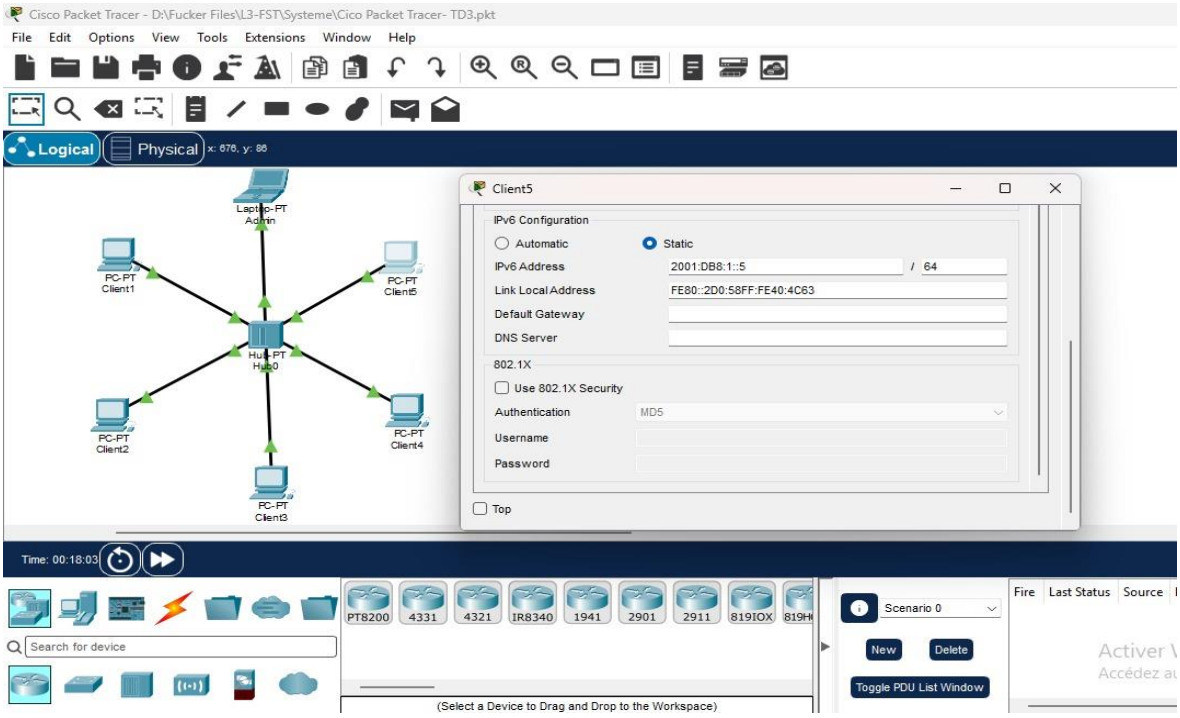
Client 4.

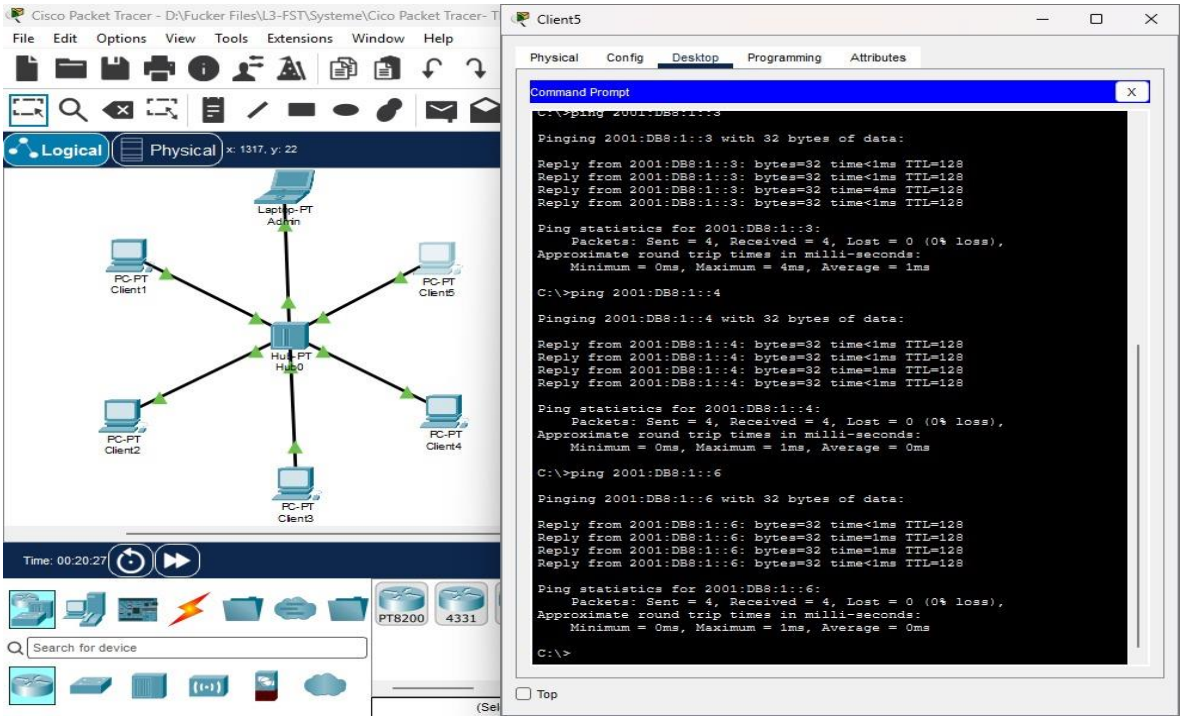

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 675, y: 33
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client4
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::4
/ 64
Link Local Address
FE80::2D0:BCFF:FE30:2256
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Time: 00:12:01

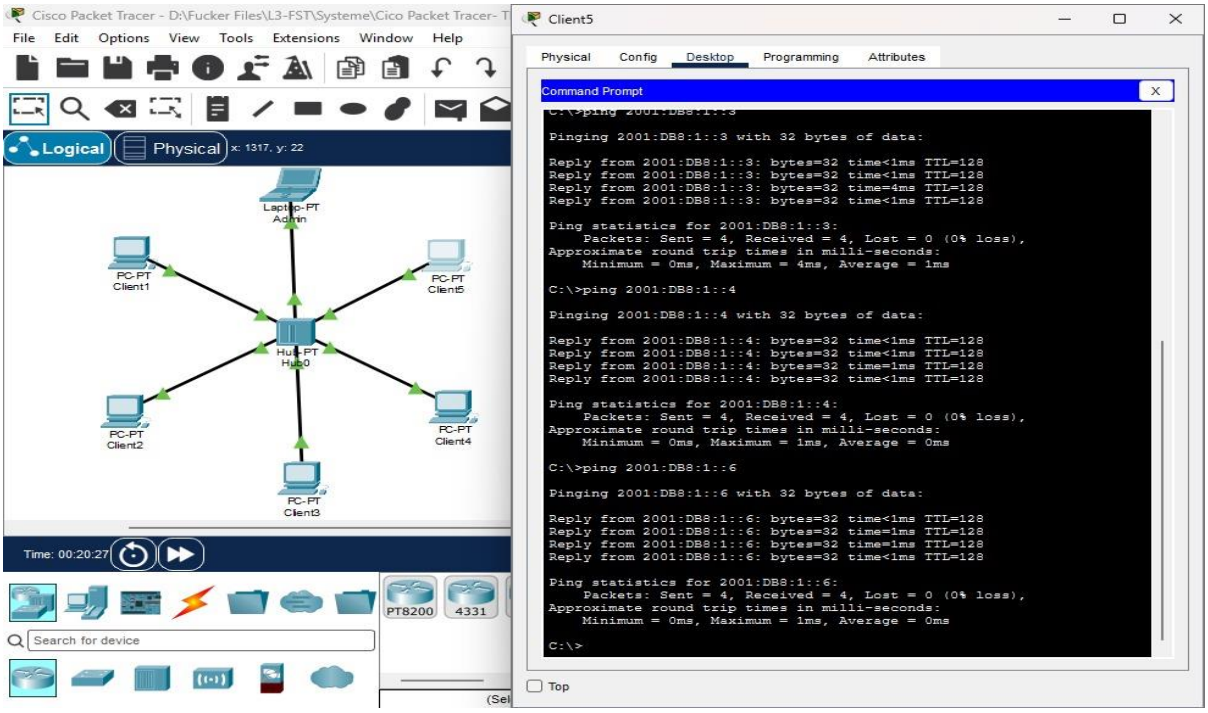

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer-
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1310, y: 47
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:16:54
PT8200
4331
Search for device
Client4
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time=5ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=9ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 9ms, Average = 2ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 4ms, Average = 1ms
Top


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer-
File Edit Options View Tools Extensions Window Help
Logical Physical x: 1312, y: 25
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:17:23
PT8200 4331
Search for device
Client4
Physical Config Desktop Programming Attributes
Command Prompt
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=4ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 4ms, Average = 1ms
C:\>ping 2001:DB8:1::5
Pinging 2001:DB8:1::5 with 32 bytes of data:
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::5: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::6
Pinging 2001:DB8:1::6 with 32 bytes of data:
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::6: bytes=32 time=1ms TTL=128
Ping statistics for 2001:DB8:1::6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top

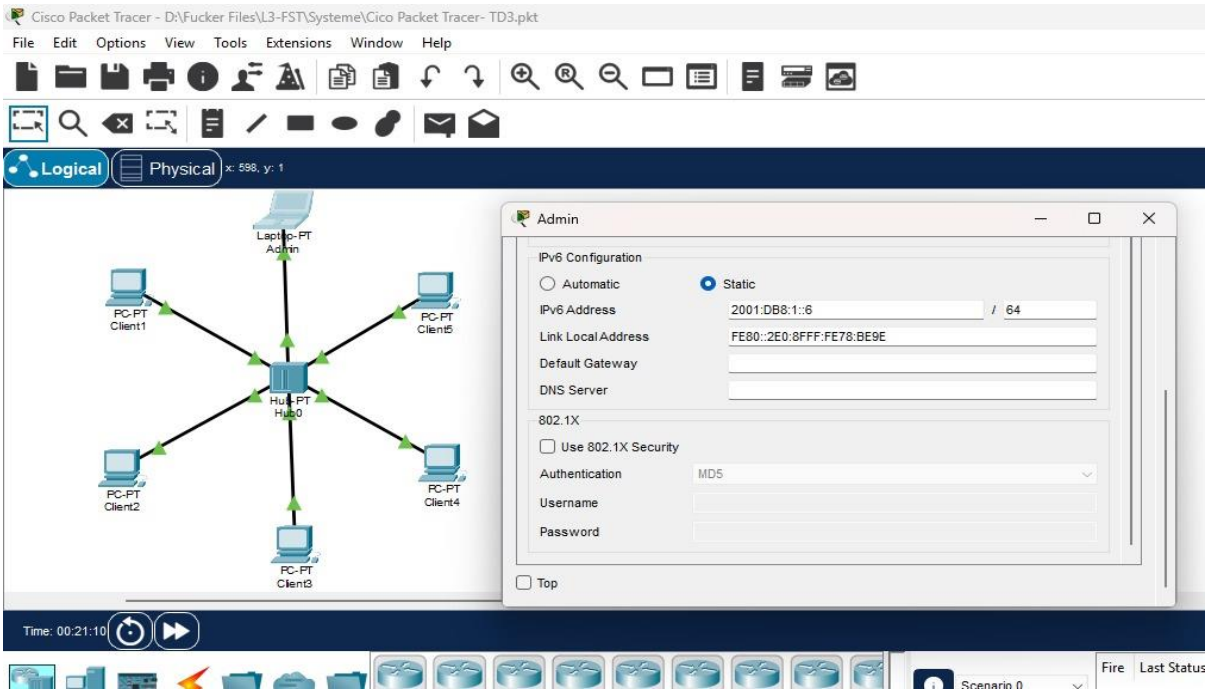
Client 5.

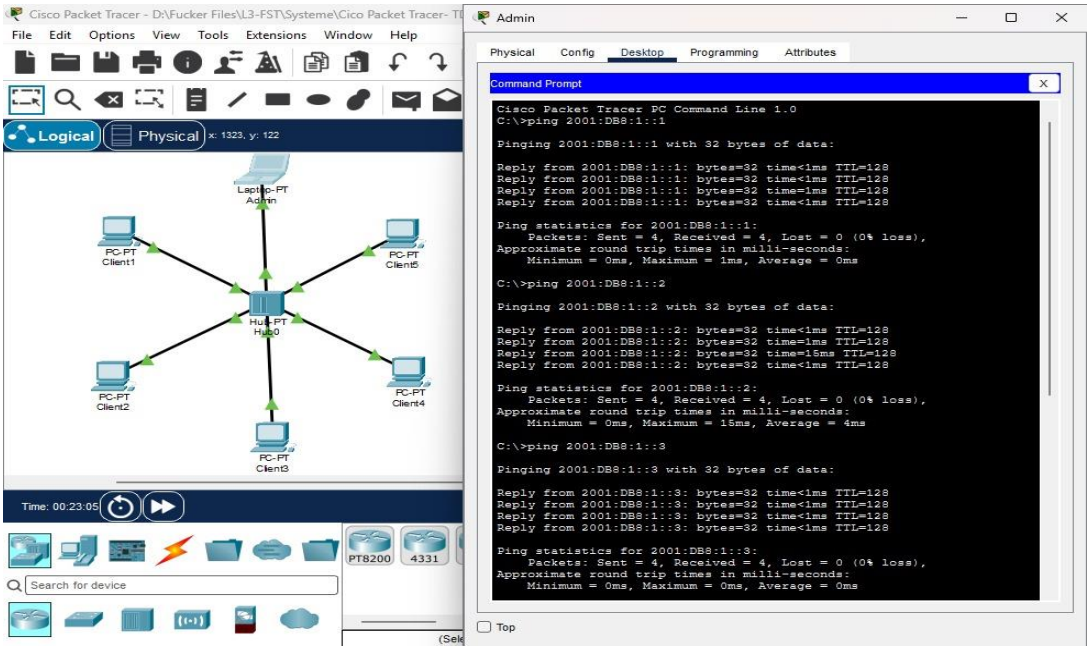

Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical Physical x: 676, y: 86
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Client5
IPv6 Configuration
Automatic
Static
IPv6 Address 2001:DB8:1::5 / 64
Link Local Address FE80::2D0:58FF:FE40:4C63
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication MD5
Username
Password
Top
Time: 00:18:03
PT8200 4331 4321 IR8340 1941 2901 2911 819IOX
Search for device
Scenario 0
New Delete
Toggle PDU List Window
Fire Last Status Source
(Select a Device to Drag and Drop to the Workspace)

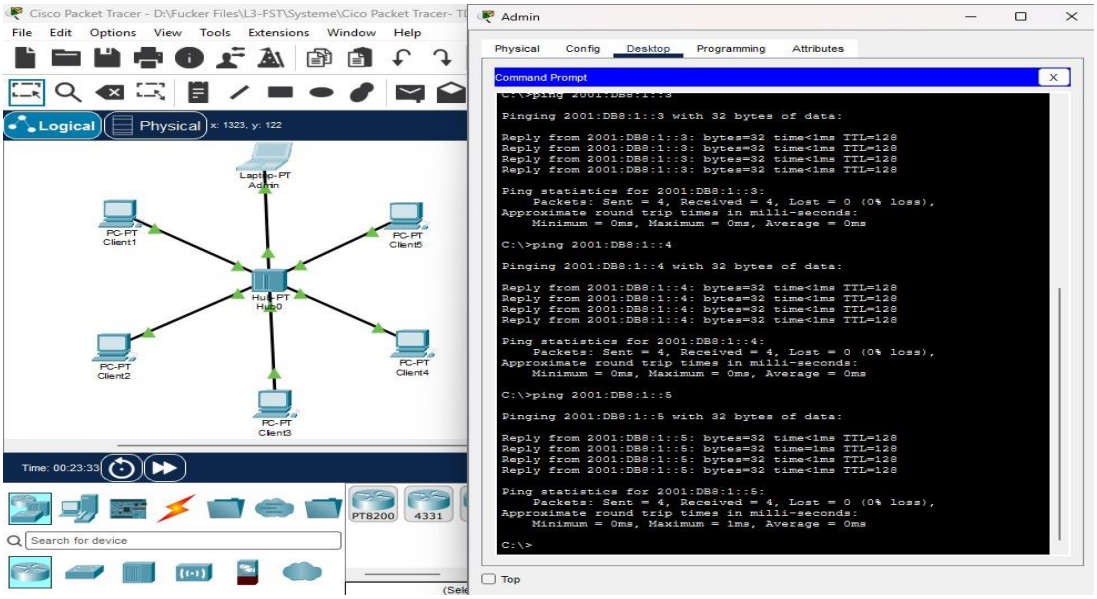




6- Configuration et vérification du Laptop Admin en IPv6 :


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 598, y: 1
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Admin
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::6
/ 64
Link Local Address
FE80::2E0:8FFF:FE78:BE9E
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Time: 00:21:10
Fire
Last Status


Cisco Packet Tracer - D:\Fucker Files\L3-FST\Systeme\Cico Packet Tracer- TD3.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1323, y: 122
Laptop-PT Admin
PC-PT Client1
PC-PT Client5
Hub-PT Hub0
PC-PT Client2
PC-PT Client4
PC-PT Client3
Time: 00:23:05
PT8200
4331
Search for device
Admin
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1::1
Pinging 2001:DB8:1::1 with 32 bytes of data:
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 2001:DB8:1::2
Pinging 2001:DB8:1::2 with 32 bytes of data:
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=1ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time=15ms TTL=128
Reply from 2001:DB8:1::2: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 15ms, Average = 4ms
C:\>ping 2001:DB8:1::3
Pinging 2001:DB8:1::3 with 32 bytes of data:
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 2001:DB8:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top



Dispositif	**IPv4**	**IPv6**
Client 1	192.168.1.10	2001:DB8:1::1/64
Client 2	192.168.1.11	2001:DB8:1::2/64
Client 3	192.168.1.12	2001:DB8:1::3/64
Client 4	192.168.1.13	2001:DB8:1::4/64
Client 5	192.168.1.14	2001:DB8:1::5/64
Admin	192.168.1.15	2001:DB8:1::6/64

Conclusion;

Ce travail dirigé m'a permis de mettre en pratique les concepts fondamentaux de l'adressage réseau à travers la simulation sur Cisco Packet Tracer.

J'ai appris à attribuer et à valider des adresses statiques tant en IPv4 (avec masque de sous-réseau) qu'en IPv6 (avec longueur de préfixe /64) sur divers équipements hôtes (PC, Laptop, Serveur). J'ai

assimilé l'utilisation des commandes ping et ping ipv6 via l'invite de commandes pour vérifier l'interconnexion globale et le bon acheminement des paquets ICMP dans le réseau.

L'expérimentation m'a permis d'observer la différence entre une topologie centralisée sur un Hub (domaine de collision partagé) et une topologie basée sur un Switch (segmentation des collisions et meilleure efficacité de communication).

Difficultés techniques et ajustements :

Durant la phase de configuration réseau sous Cisco Packet Tracer (TD3), nous avons confronté un rejet systématique du bloc d'adresses 3000:is9:1::X. Une vérification de la norme IPv6 nous a permis de constater que seuls les symboles hexadécimaux (0-9 et A-F) sont admis. L'invalidation provenait de la présence du caractère non conforme « s ». Pour remédier à cette anomalie et mener la simulation à terme, nous avons révisé notre plan d'adressage en optant pour le sous-réseau réglementaire.